Vectorization Method for Remote Sensing Object Segmentation Based on Frame Field Learning: A Case Study of Greenhouses

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***Abstract*— Deep learning technologies have significantly advanced object information extraction from remote sensing data in recent years, achieving broad application across various industrial sectors. However, information loss exists between remote sensing object raster segmentation and geographic information vector mapping, making it challenging to directly apply raster extraction results to vector mapping. This study, taking the automatic extraction of greenhouses based on remote sensing imagery as an example, proposes a vectorization method for remote sensing object segmentation based on frame field. This method bridges the gap between the object pixel segmentation process and the mask vectorization process through the frame field information outputted by the network, resulting in smoother and more regular vector extraction results. To validate the effectiveness of our framework, we introduce the first high-precision greenhouse vector boundary dataset. Extensive experiments demonstrate that our method significantly mitigates the information loss issue prevalent in traditional vectorization processes, achieving a 5.05% improvement in intersection over union (IoU), a 6.06% increase in recall, and a 5.54% reduction in maximum angular error (MAE) compared to simple vectorization schemes. It outputs more regular greenhouse vector plots, where the precision of the frame field plays a crucial role in the final vectorization quality. This research offers a unique and practical solution, converting remote sensing object segmentation into vector maps.**

***Index Terms*— Activate contour model, deep learning, frame field learning, greenhouse.**

I. Introduction

THE process of accurately marking objects in an image at the pixel level is known as [1]. Currently, deep learning has been widely applied to object segmentation due to its powerful capability to represent features [2], [3], [4], [5]. This has significantly enhanced the efficiency and precision of extracting features such as buildings [6], [7], roads [8], and farmlands [9] from high-resolution satellite imagery. Deep learning has become the mainstream method for rapid object extraction, contributing to some extent to the modernization of natural resource surveys, land resource mapping, and other related fields. Especially in recent years, the emergence of remote sensing pretrained models like SpectralGPT [10] and Skyscence [11], through unsupervised learning on massive amounts of remote sensing data, has broken the model's dependence on labeled data, significantly enhancing the accuracy and generalization ability of these models in image segmentation tasks [12].

However, natural image segmentation only partially meets real-world geographic surveying and mapping needs. In natural image segmentation, the focus is on identifying the overall shapes of objects, while geographic surveying and mapping require defining the accurate vector boundaries between different land features. Existing methods [13] in interpretation and mapping adopt a two-step approach: "interpretation first, vectorization second." This process includes: 1) creating a binary raster map; 2) transforming the raster map into vector polygons; and 3) streamlining the initial polygons using a fixed-parameters method [14], [15] to cut down on unnecessary vertices. The accuracy of the final vectorized results for the target objects largely depends on the quality of the binary mask output by the model. Although the segmentation precision of binary masks has significantly improved with advancements in deep learning technology [16], we have observed that it is challenging to directly restore the original vector annotation form from the binary results produced by deep learning models. However, as long as the resolution remains within a reasonable range, the conversion process between the ground truth (GT) and the annotations is relatively straightforward. This issue is referred to as *mask reversibility* in [17], and obtaining this reversible information directly through deep learning is notably challenging. Furthermore, the phenomenon tends to be more evident in objects with more regularized boundaries but subtler gradients in image features, such as greenhouses.

Greenhouses are crucial in modern agriculture, contributing immeasurably to increased agricultural production and income [18]. The lifespan of greenhouses varies significantly, and they can change rapidly. Therefore, it is essential to

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precisely obtain spatial distribution information of greenhouses from remote sensing data [19], [20], [21], [22]. Greenhouses come in various types, but most have regular geometric shapes with semi-transparent film covering their tops. These characteristics make them reasonably straightforward to detect in remote sensing imagery. Nevertheless, the imaging angle, environmental conditions, and light reflection from internal crops and adjacent ground objects can obscure the gradient information at their boundaries, making these features less pronounced than in other objects [23]. Furthermore, compared to other artificial structures, greenhouses are often situated very close to each other. This proximity can result in the loss of boundary information as early as the input mask is generated, especially if traditional sample preparation methods are employed. Therefore, although deep learning techniques have achieved very high accuracy in obtaining greenhouse binary labels from remote sensing imagery [22], the practical application of fine mapping for greenhouses still needs to be improved. Due to semantic ambiguity in the edge pixels, the segmentation mask lacks reversibility (mask-reversibility). This results in jagged and irregular edges in the vectorized greenhouse contours, which do not meet modern agriculture's practical requirements for precise management.

Numerous methods have been proposed to bridge the information gap between remote sensing image segmentation and object vector mapping. These can be mainly divided into two processes: 1) from the perspective of information extraction, the aim is to enhance the network's ability to represent edge information of objects. This can be achieved by using loss functions or network structures that emphasize edge information, guiding the network to focus on edges while downplaying internal information [24], [25] and 2) from the perspective of post-processing, the goal is to improve the binary mask vectorization process by removing as many redundant inflection points as possible [26], [27]. The Douglas-Peucker (DP) algorithm [14] and the contour tracing algorithm [28] are classic techniques used for vectorization. However, these methods have their drawbacks. They require the manual setting of thresholds, and inaccuracies can arise if these thresholds are not correctly adjusted, especially when there is a significant size variation between objects. Therefore, if these two processes could be unified, or if the parameters for the latter could be adaptively driven by the former, it would be highly beneficial for the fine mapping of greenhouses.

Unfortunately, in numerous previous studies, image segmentation and object vectorization processes have been disjointed. This separation is partly due to the necessity of adapting to the paradigms of deep learning applications and partly due to the inability of traditional architectures to model geometric structures effectively. To integrate these two processes, numerous scholars have made efforts. For example, some have attempted to guide the vectorization process by outputting structured information through a raster, effectively connecting the two processes. A notable example is the frame field [29], which has demonstrated superior performance. A frame field is a matrix the same size as the image, containing values representing various information pieces, such as the tangent direction and the rate of change at each point in the image. This technique has been widely applied in numerous domains, including surface parameterization, texture mapping, DEM generation (rendering), image preprocessing, and more. The frame field algorithm has been successfully applied to building segmentation, achieving promising results. Nevertheless, given the spectral diversity of greenhouses and the impact of mixed pixels at their boundaries, whether this algorithm can be transferred to greenhouse mapping remains to be investigated.

To bridge the gap in information representation between binary segmentation of remote sensing imagery and precise vector mapping of greenhouses, we have combined a deep learning-based binary segmentation model for greenhouses with a prior knowledge-driven raster interpretation result vectorization process, creating a technical framework that directly outputs high-precision vector extraction results from remote sensing images. The main contributions of this study include as follows.

1) *Multioutput Neural Networks Incorporating Frame-Field Learning:* We have developed a three-task neural network to enhance the precision of image edge segmentation while simultaneously outputting frame-field information about the direction of edge gradient changes. This information serves as prior knowledge for the vectorization process, aiding in the accurate drawing of vector lines.
2) *Automatic Vectorization Process Driven by Multiple Energy Functions:* We introduced a geoscientific active contour model (GACM) to enhance the vectorization process. It utilizes the frame field and boundary information output from the three-task neural network as internal and external energy functions, driving the automated vectorization process and incorporating a series of expert knowledge to achieve high-precision greenhouse mapping.

These methods effectively build a bridge between the interpretation and vectorization processes, avoiding the accumulation of errors in format conversion, and significantly enhancing the precision of greenhouse vector mapping without adding excessive parameters. This framework not only improves the accuracy of greenhouse identification and mapping but also provides a reliable reference model for similar remote sensing applications.

II. Datasets

A. Study Area

Shouguang is located in the northeastern part of Weifang City, Shandong, adjacent to the Bohai Sea. As of 2021, Shouguang has more than 600 000 acres of agricultural greenhouses, accounting for over one-fourth of the city's total land area. It is rightfully known as the "Hometown of Greenhouses" and serves as the "Vegetable Basket" for the entire nation, playing a crucial role as a significant demonstration base for modern facility agriculture in China. In this study, we utilized remote sensing imagery from the Super-view-1 (SV1) satellite, captured in the year 2020, which includes mosaic images from all four seasons: spring, summer, autumn, and winter. The satellite data has four multispectral bands, each with a spatial resolution of 2.0 m and one panchromatic band with a spatial resolution of 0.5 m. After pan-sharpening, we can obtain four bands with an enhanced spatial resolution of 0.5 m.

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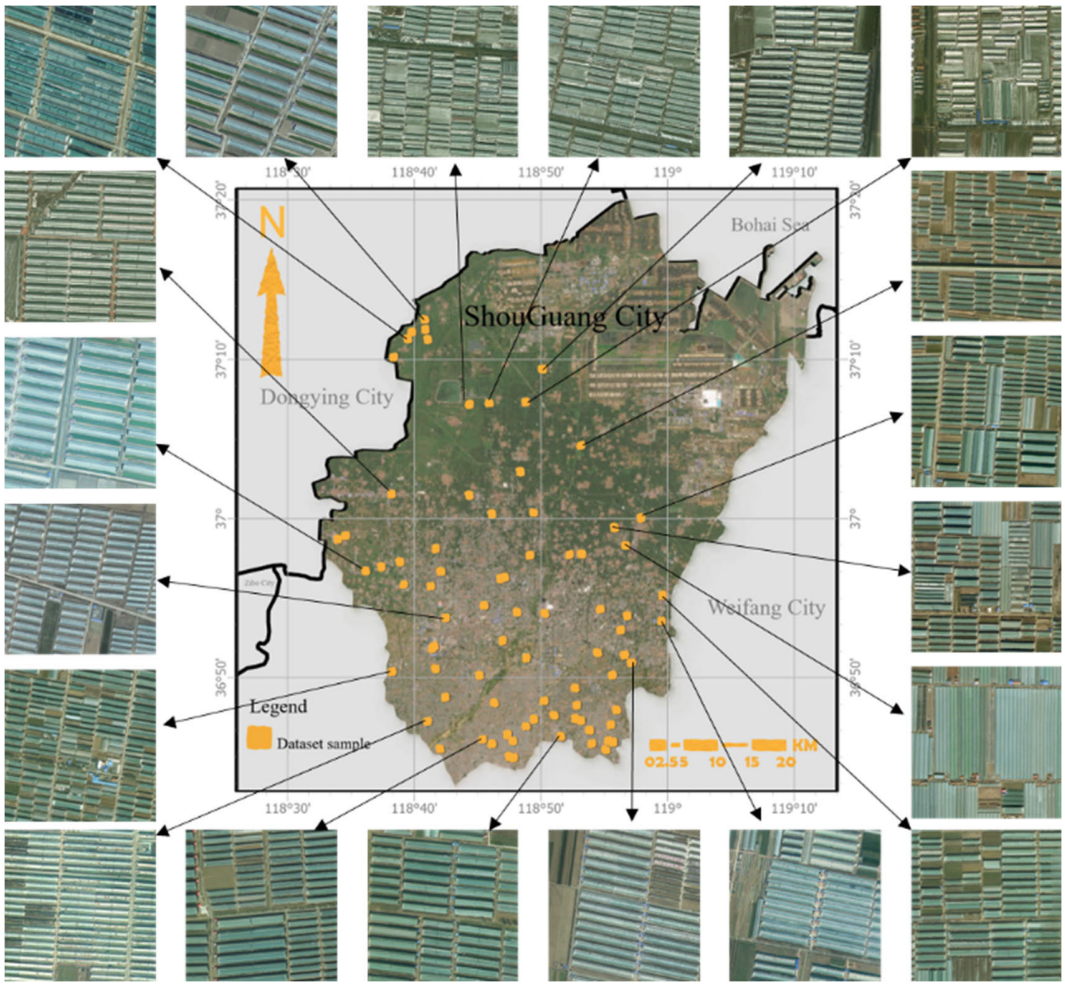


Fig. 1. Schematic representation of some of the images in the dataset and the distribution of all samples. The spatial resolution of the image is 0.5 m and the size is 1024 × 1024 pixels.

The SV1 satellite boasts the highest spatial resolution among civilian satellites in China, making it a crucial data source for geographical national condition monitoring, nonagricultural and nongrain monitoring, and other related tasks.

B. Datasets

Shouguang, with its numerous, densely distributed greenhouses varying in size, structure, material, and purpose, serves as an exemplary region for conducting greenhouse remote sensing extraction studies. Within the study area, we have randomly selected regions of size 1024 × 1024 for manual annotation of greenhouses. To meet the requirements of precise mapping, in this study, all individual greenhouses are delineated as separate objects based on their boundaries, avoiding any shared edges even where greenhouses are situated extremely close. To guarantee accuracy, the annotation process is independently completed by two individuals, followed by a thorough cross-verification to eliminate any discrepancies. The final dataset comprises remote sensing imagery paired with corresponding vector annotation labels (formatted as shapefiles), covering a total area of 2072 km^2. This dataset includes various land features, including farmland, greenhouses, and residential areas. The spatial distribution of the samples is illustrated in Fig. 1. To facilitate imagery input into the neural network model, the aforementioned annotated dataset is cropped into tiles of size 512 × 512, resulting in a total of 2531 images. Of these, 2025 are used for training, while the remaining 506 are reserved for testing. An illustrative example of a data pair is shown in Fig. 2. Traditional remote sensing image segmentation involves converting the vector labels [Fig. 2(b)] directly into a binary image [Fig. 2(c)], which shares the same range and resolution as the raster image. This meticulous data preparation and annotation approach ensures a comprehensive and accurate dataset ready for the following advanced analysis and modeling.

C. Frame Field Labels

To emphasize the significance of edge information, we have produced additional binary edge labels [Fig. 2(d)] and frame field labels [Fig. 2(e)], going a step further than merely utilizing binary labels. The binary edge labels are derived by transforming the vector labels into linear vectors, whereas the frame field labels are computed utilizing the greenhouse vector map.

The process of calculating the frame field labels is as follows: First, we compute each edge's angle (in radians) in

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(a) (b) (c) (d) (e)

Fig. 2. Schematic of a dataset pair composition. (a) Remote sensing image, (b) vector boundary label of the greenhouse, (c) binary image label, (d) binary edge label, and (e) frame field label. The frame field labels are obtained by calculating the angle of each boundary based on the vector boundary label.

the vector map

$$\theta' = \begin{cases} \theta, \theta > 0 \\ \theta + \pi, \theta < 0 \end{cases} \tag{1}$$

where θ represents the actual angle of the edge. Subsequently, we map this angle to the range of 0–255 and convert the vector map into a raster frame field label that matches the resolution of the remote sensing image [Fig. 2(e)]. In this raster representation, the pixel values at positions corresponding to the greenhouse boundaries represent the angle values of those boundaries. For positions that are far from the greenhouse boundaries, we set their values to 0.

The inclusion of these specialized labels ensures that our approach has access to crucial information about the boundaries and orientations of the greenhouses, facilitating a more accurate and nuanced understanding of the spatial relationships within the imagery.

III. Methods

To fully leverage the powerful fitting capabilities of deep learning, this study introduces a greenhouse vectorization framework based on frame field learning, as illustrated in Fig. 3. The framework includes the following three steps. First, a relationship between remote sensing images and label datasets is established using a neural network model, enabling the extraction of greenhouse binary maps, edge maps, and frame field maps from remote sensing images. Subsequently, morphological operations and vectorization processes are applied to the greenhouse binary map to obtain initial vector data. Finally, the greenhouse edge map, vectorized greenhouse binary map, and greenhouse frame field map are input into the GACM. Through algorithmic iterative optimization, the optimized greenhouse vector labels are obtained.

A. Frame Field Learning

The employment of the frame field as a guiding mechanism for the greenhouse vectorization procedure is a crucial component of this research. The frame field constitutes a spatial domain encapsulating critical information, including the tangent direction and variation intensity at each point across the image. Inspired by [30], this experiment has constructed a vectorization framework. Utilizing the frame field facilitates the optimization of greenhouse contours post-vectorization, culminating in representations of greenhouse structures that are both more regular and precise.

To align with the deep learning paradigm, we draw inspiration from [27] and adopt a complex number representation for the sine and cosine values of the angles formed by tangents or normal on the greenhouse contours. As a result, the frame field can be formulated as follows:

$$z = \sin(\theta')i + \cos(\theta') \tag{2}$$

for the frame field z in the labels, we use the components u and v in two distinct directions, respectively, to approximate the true values during the neural network training process. For each edge on the contour, two variables are defined to represent the tangent and normal directions at each point on the greenhouse contour. If the point is a corner point on the contour, it represents the direction from the corner point to the next and the previous corner points. Therefore, for each z on the contour, there always exists a relationship that can be expressed as

$$f(z) = (z^2 - u^2)(z^2 - v^2). \tag{3}$$

In the equation, when u and v, respectively represent the tangent and normal vectors, $f(z) = 0$. Therefore, by accurately obtaining the values of u and v, we can derive z, and subsequently calculate the angle of the greenhouse contour θ'. However, when either u^2 or v^2 is equal z^2, $f(z) = 0$, making it impossible to calculate the value in the other direction using this function. To circumvent this issue, Girard et al. provide a solution [29]: instead of directly obtaining u and v, the formula is expanded to

$$f(z) = z^4 + c_2 z^2 + c_0 \tag{4}$$

where,

$$\begin{aligned} c_0 &= u^2 v^2 \\ c_2 &= -(u^2 + v^2). \end{aligned} \tag{5}$$

By calculating c_0 and c_2 through the network, we can indirectly obtain the values of u^2 and v^2, which in turn enables us to determine the direction of the edge lines in the greenhouse labels

$$\begin{aligned} u^2 &= -\frac{1}{2}\left(c_2 + \sqrt{c_2^2 - 4c_0}\right) \\ v^2 &= -\frac{1}{2}\left(c_2 - \sqrt{c_2^2 - 4c_0}\right). \end{aligned} \tag{6}$$

This is crucial for the vectorization process, as it enhances the direction of the representation of the greenhouse contours.

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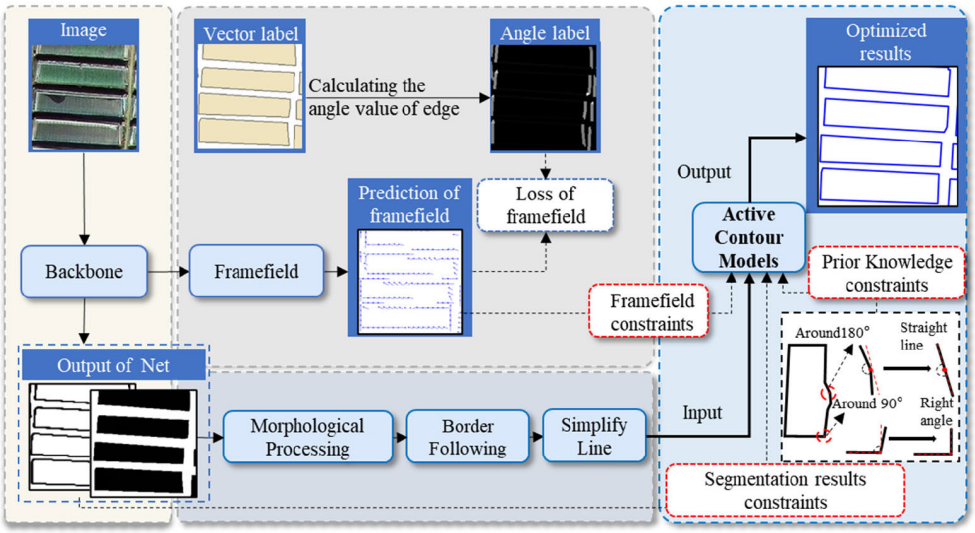


Fig. 3. Schematic of the greenhouse extraction process proposed in this study. The process is divided into four parts (a) obtaining the segmentation and edge maps of the greenhouse using neural networks, (b) obtaining the frame field map of the greenhouse using the frame field's prediction header, (c) initial vectorization of the segmentation maps of the greenhouse using a simple vectorization process, and (d) optimization of the greenhouse vectors using the ACM.

B. Three-Task Neural Network

To meet the requirements of our task, we have designed a multitask end-to-end neural network capable of simultaneously extracting greenhouse binary maps, greenhouse edge maps, and greenhouse frame fields. As illustrated in Fig. 4, the network is based on the U-Net architecture, embodying an encoder-decoder structure tailored for segmentation tasks. The decoder part is divided into two main components: one for semantic segmentation and the other for edge detection. The semantic segmentation component is responsible for capturing the contextual information of the image. The edge feature extraction employs the convolutional block attention module (CBAM) [31] to suppress information related to nongreenhouse edges in the image, thereby directing focus toward the greenhouse edges.

The network's encoder utilizes BoT-Net [32], a sophisticated model that integrates transformers with convolutional layers to capture the hierarchical features of greenhouses in remote-sensing images. The decoder part utilizes several up-sampling modules to enlarge the feature maps while reducing the channels, outputting high-level features of the greenhouse. In the final phase of the process, these high-level features, encapsulating information about the greenhouse and its edges, are fed into the spatial pyramid pooling (SPP) model [2]. In this setup, we use two SPP models with nonshared parameters: one outputs the binary segmentation map of the greenhouse, and the other yields c_0 and c_2 after processing through the Tanh activation function, thus obtains the frame field of the greenhouse. Additionally, the high-level features of the greenhouse edges obtained from the edge extraction part are upsampled and convolved to output the greenhouse edge map. Such a design significantly reduces the number of parameters in the network, enhancing the network's operational speed and efficiency.

C. Loss Function

We have adopted a composite loss function L_{com} to guide the training of the three-task network better. As obtaining the greenhouse binary map and edge map can both be considered pixel segmentation tasks, we have employed the same loss function, specifically a combination of binary cross entropy (BCE) and dice loss functions. This combined loss function has been proven to be extremely effective in various binary segmentation tasks involving neural networks [7], [33]. It can ensure a balanced and effective training process, optimizing the network's performance in segmenting the greenhouse structures and their edges [34]. The formula for this loss function is as follows:

$$L_{\text{BCE}} = -(y \log(p(x)) + (1-y)\log(1-p(x))) \tag{7}$$

$$L_{\text{Dice}} = 1 - \frac{2p(x)y}{p(x)^2 + y^2} \tag{8}$$

$$L_{\text{Seg}} = L_{\text{BCE}} + L_{\text{Dice}} \tag{9}$$

where y represents the GT values and $p(x)$ represents the values predicted by the model.

The loss function plays a pivotal role in guiding the greenhouse frame field model toward effective performance. A loss function designed for predicting the frame field using neural networks was initially proposed in [29]. However, its complexity poses challenges in achieving convergence during the training phase. To address this issue, this study strategically omits certain components of that loss function that exert a negligible influence on the frame field prediction, streamlining the process. The comprehensive formulation of the frame field loss function is depicted in (3)–(6). The overall construction of the frame field loss function is depicted in (3)–(6). The network outputs c_0 and c_2 and utilizing (6), the value of θ is subsequently determined.

Furthermore, to expedite the convergence of the frame field, and inspired by [35]. we delineate several optimization rules specific to the frame field. First, when a pixel is located at the contour edge, it is imperative that at least one of u and v aligns with the tangent direction of the greenhouse contour, thereby describing the general trend of the greenhouse edge. Second, when a pixel is located at the contour corner, the vectors u and v should each point toward the tangent

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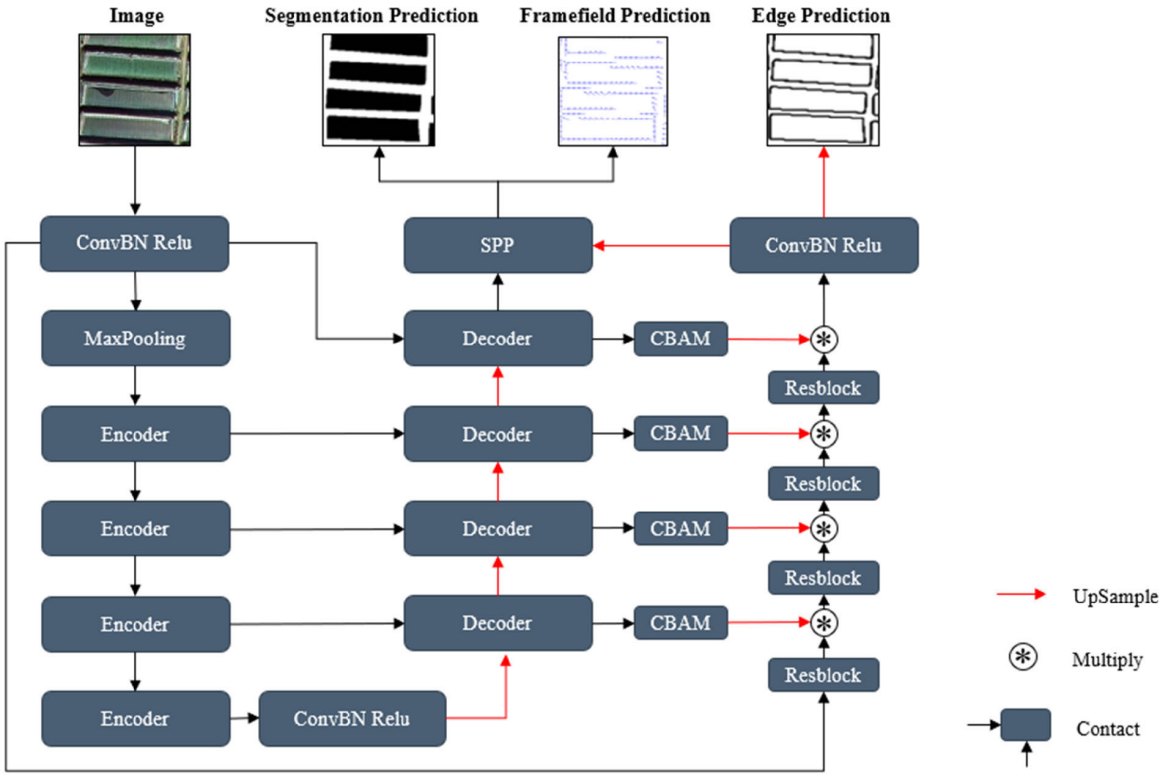


Fig. 4. Network architecture employed in this study, is used for the extraction of the greenhouse segmentation binary maps, edge maps, and frame field maps.

directions of the two edges converging at the corner, effectively encapsulating information about the greenhouse's angular features. Finally, for vectors situated at a distance from the greenhouse, maintaining smoothness is key to preventing mode collapse during training, which could otherwise culminate in the vectors converging to form a straight line. To ensure that the greenhouse frame field does not misguide the greenhouse contour in subsequent iterations, we apply the Laplace operator to smooth the frame field at positions far from the greenhouse edge. The culmination of these efforts results in a loss function adept at predicting the frame field using neural networks, succinctly expressed in the following equations:

$$\begin{aligned} L_{frame} = \min_{c_0,c_2:I\to C}(&\lambda_0 \mathrm{int}_I \left| f(z; c_0(\vec{x}), c_2(\vec{x}) \right|^2 d\vec{x} \\ &+ \lambda_1 \mathrm{int}_I \left| f(z^{\perp}; c_0(\vec{x}), c_2(\vec{x}) \right|^2 d\vec{x} \\ &+ \lambda_2 \sum_{i=0,2} \mathrm{int}_I \left|\left| \nabla c_i(\vec{x}) \right|\right|^2 d\vec{x} \\ &+ \lambda_3 \mathrm{int}_I \left| f(\nabla y_{\mathrm{int}}; c_0(\vec{x}), c_2(\vec{x}) \right|^2 d\vec{x}) \end{aligned} \tag{10}$$

where, I represents the set of all greenhouse edges. c_0 and c_2 are regarded as functions of the pixel position $\vec{x}$, z is the complex form of the greenhouse frame field label, $z^{\perp}$ is the perpendicular vector of z, and ∇y_{int} is the gradient map calculated based on the predicted binary map of the greenhouse.

D. Geoscientific Active Contour Model

Active contour models (ACMs) [36], also known as snakes, are a widely used technique in image processing and computer vision for object segmentation and boundary extraction. By iteratively adjusting an initial contour under the influence of internal and external loss functions, ACM ensures that the final contour is both smooth (due to internal loss function) and accurately aligned with the target object's boundaries (due to external loss function). This dual-energy approach enables ACM to effectively handle variations in object shape and appearance, making it a robust tool for various applications such as medical image analysis, object tracking, and computer-aided design. In this study, we integrate ACM with a focus on greenhouse extraction tasks, proposing GACM. GACM treats the frame field map as an internal loss function $\mathrm{En}_{\mathrm{Int}}$ and combines it with an angle internal energy constructed based on prior knowledge to drive the changes in the greenhouse vector contour. Simultaneously, the greenhouse edge map predicted by the neural network is used as an external loss function $\mathrm{En}_{\mathrm{Ext}}$ to constrain the changes in the greenhouse vector contour. Through iterative algorithmic adjustments, a smooth and regular greenhouse vector contour is ultimately obtained. Compared to traditional ACM, GACM places a greater emphasis on the application of geoscientific features while also incorporating the outputs of neural networks. This approach enables the achievement of more regular and smoother greenhouse vector contours.

The specific process is illustrated in Fig. 3. First, we utilize a neural network to obtain the vectorized result of the greenhouse binary map as the initial contour. Before vectorization, we perform erosion and dilation on the greenhouse binary image to reduce the computational load during the vectorization process. Second, GACM uses the frame field map obtained from the neural network and the tangent direction

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of the vectorized greenhouse contour for calculations, thereby driving the edges of the greenhouse to rotate in the direction of the frame field. At the same time, we define that the angles in the vector edges should approximate either 90° or 180°. The former constrains the corners of the greenhouse, while the latter smoothens the edges of the greenhouse. These two loss functions act as internal loss functions $\mathrm{En}_{\mathrm{Int}}$, driving the changes in the greenhouse edges. Finally, to restrict drastic changes in the greenhouse edges, we have proposed an external loss function $\mathrm{En}_{\mathrm{Ext}}$—the predicted greenhouse edge map. This ensures that the changes in the greenhouse edges remain as close as possible to the predicted greenhouse edge map, preventing them from straying too far from the actual greenhouse edges. The process continues until there is a good alignment between the vectorized greenhouse data, the predicted greenhouse edge, the greenhouse frame field, and the angles.

IV. Result

A. Experimental Design and Evaluation Metrics

This research is fundamentally segmented into two main components: the extraction of segmentation results for greenhouses using a three-task neural network, and the vectorization of these results facilitated by the GACM. To evaluate the performance of the outputs at each respective stage, comparative experiments have been meticulously crafted and are expounded upon in Sections IV–II and IV–III, aimed at demonstrating the efficacy of our proposed methodology. In the segmentation phase, we have selected to utilize Deeplabv3+ and U-Net as benchmark architectures for comparative analysis. Both architectures are renowned in the domain of image segmentation and have proven their mettle across diverse application areas. In the subsequent vectorization stage, our results obtained through the GACM methodology are contrasted against those derived from a simplistic vectorization approach as well as the results yielded by the DP algorithm. This comparative analysis is underpinned by a series of meticulously selected metrics, facilitating both qualitative and quantitative evaluations of the respective methodologies.

In this research, a suite of five distinct metrics has been selected to comprehensively assess the performance of both the neural network model and the GACM. The metrics encompass overall accuracy (OA), recall, intersection over union (IoU), the maximum angular error (MAE) [29], and the $1/f(z)$ ($\mathrm{F}_{\mathrm{error}}$) metric. For the evaluation of binary greenhouse images, we have opted to engage in a pixel-wise statistical analysis, quantifying the true negative (TN), false negative (FN), true positive (TP), and false positive (FP) pixels. Conversely, in appraising vectorized greenhouse images, we proceed to calculate the areas corresponding to TN, FN, TP, and FP.

The definitions for each of the metrics are articulated as follows:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{11}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \tag{12}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}. \tag{13}$$

The neural network's prediction is oriented toward estimating the frame field parameters c_0 and c_2. Given that additional loss functions are incorporated during the prediction phase, (3) serves as a viable metric to evaluate the precision of the frame field predictions. Equation (14) involves taking the reciprocal of $f(z)$. The $1/f(z)$ criterion provides a more intuitive means of discerning the quality and reliability of the predicted frame field outcomes. A larger value of F_{error} indicates higher accuracy of the frame field. This provides a more intuitive method for assessing the quality and reliability of predicted frame field results

$$F_{\mathrm{error}} = \frac{1}{f(z)} \tag{14}$$

where, the specific definition of $f(z)$ can be found in (3). The MAE is a metric to assess the extent of directional deviation between two vector entities. It involves partitioning one of the vector entities into V smaller line segments, subsequently projecting these segments onto the other vector. The ensuing step entails calculating the angular values between the vector onto which the segments have been projected and the projected segments. The MAE is defined as the largest angular value among all these small segments. The formula for calculating the MAE is presented as follows:

$$\mathrm{MAE} = \max_{j \epsilon V} \Delta\theta_j \tag{15}$$

where V represents the set of all the small line segments and $\Delta\theta$ denotes the angle between each projected small line segment and the other vector entity.

B. Segmentation Performance of the Greenhouse

To validate the three-task network's effectiveness, we compared various network architectures using our dataset. Fig. 5 shows that our neural network model quickly reduced training loss at the start. As training continued, our model's validation loss stabilized and remained on par with U-Net's, ultimately converging to a similar final value. However, Deeplabv3+ underperformed in comparison, struggling more with training loss and validation loss. This suggests it might not be as suitable for binary greenhouse image extraction, possibly due to limitations in its boundary representation components. In summary, our model converges quicker initially and matches UNet in overall performance.

After completing model training, we assessed the performance of various networks using the test dataset. Our neural network model significantly outperforms Deeplabv3+ across OA, IoU, and Recall metrics (Table I) and also showed marked improvements over the U-Net model. Specifically, in extracting greenhouse frame fields, our approach demonstrates superior accuracy compared to the U-Net and Deeplabv3+ models. This improvement can be credited to BotNet's multihead self-attention mechanism, which separately handles frame field information. By doing so, it avoids potential interference between the frame field data, edge details, and their respective mask data, ensuring not only accurate greenhouse mask extraction but also a higher precision in frame field information output.

It needs to be emphasized that the precision of the segmentation mask and the accuracy of the frame field are both crucial,

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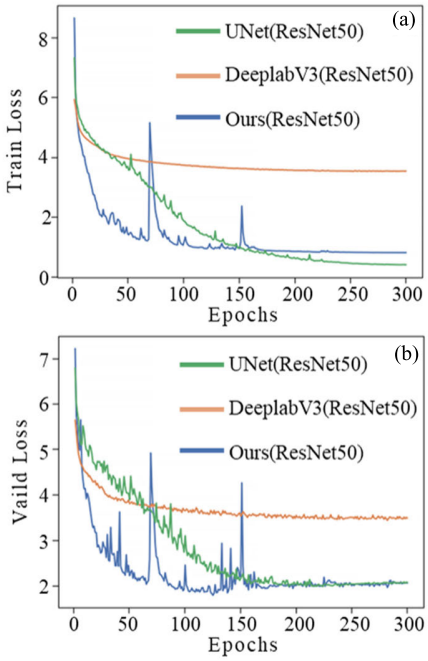


Fig. 5. Loss function curves of different models (a) training and (b) validation.

TABLE I
PERFORMANCE COMPARISON ON THE TEST DATASET AMONG DIFFERENT SEGMENTATION NEURAL NETWORKS

	IoU	Recall	Accuracy	F_{error}
U-Net	0.9394	0.9720	0.9712	133.33
Deeplabv3+	0.6527	0.8920	0.7888	87.72
Ours	0.9579	0.9776	0.9800	217.39

as they play a key role in determining the final extraction results of the greenhouse. In this study, our primary focus is on achieving a higher precision in the vectorization process based on raster extraction results, which holds true even when operating within more universally applicable network frameworks. Consequently, the impact of complex networks on the model's segmentation precision is not within the scope of our current discussion and is not explored in this study.

Table II showcases the parameters and average training time per epoch for the tri-task neural network model. In this network, the baseline outputs segmentation objects, edges, and their frame fields simultaneously, whereas the "Without framefield" configuration degenerates to a standard segmentation network, only outputting segmentation results and edges, without involving flow field prediction. Through comparison, we observe that introducing the flow field prediction module into the model has a relatively minor impact: it only increases the parameter count by 0.02% and the training time per epoch by 0.15%. This indicates that although the flow field prediction adds additional computational demands, its impact on the overall model size and training efficiency is very limited.

C. Performance of Greenhouse Extraction Vectorization Results

The binary segmentation maps, edge maps, and frame field maps generated by the aforementioned neural network model

TABLE II
TIME AND SPACE COMPLEXITY OF OUR NEURAL NETWORKS

	Baseline	**Without framefield**
Parameters	78698776	78677236
Time (one epoch)	164.50 s	164.26 s

TABLE III
COMPARISON OF THE ACCURACY OF GREENHOUSE VECTOR MAPS OBTAINED VIA DIFFERENT VECTORIZATION METHODS

	IoU	Recall	MAE
Simplistic	0.7185	0.7215	1.5544
DP	0.7180	0.7210	1.5462
GACM	0.7690	0.7821	1.4682

serve as inputs for the GACM, facilitating the acquisition of vector-formatted results for greenhouse extraction. As a comparison, we elected to implement both the simplistic vectorization method and the DP vectorization method. Simple vectorization involves vectorizing the binary image of the greenhouse outputted by the neural network along the grid's edges. In contrast, the DP method entails the application of the DP algorithm to simplify the results derived from simple vectorization. Table III displays IoU, Recall, and MAE associated with the various vectorization methods. The vectorization results for selected typical greenhouses are illustrated in Fig. 6.

The GACM employed in this study, guided by frame fields and expert experience, demonstrated a relative enhancement in performance compared to the simple vectorization method that undergoes no further processing. Specifically, there was a 5.05% increase in IoU, a 6.06% augmentation in Recall, and a 5.54% decrease in the MAE. Conversely, examining the classic DP method, the IoU and recall exhibit a slight decline compared to the simple vectorization method. The MAE metric shows improvement, which only decreased by a modest 0.53%. It is imperative to note that the DP algorithm achieves edge smoothing through threshold comparison, with this threshold set at 0.8 in our study, corresponding to half the diagonal length of a pixel. However, as can be seen from Fig. 6, the DP algorithm's performance is contingent upon the choice of threshold, enabling it to eliminate certain jagged edges. In regions where the greenhouse contours are particularly curvaceous, the DP algorithm exhibits suboptimal performance and may even lead to information loss.

The GACM is adept at modifying the trajectory of the greenhouse's vector boundaries by the field information provided by the frame field, thus fulfilling the objective of smoothing jagged edges. Notably, this process does not necessitate the manual setting of a threshold, showcasing GACM's ability to optimize vector boundaries autonomously.

The greenhouse contours outputted by the GACM are smoother than other compared methods. While effectively removing the contour jaggedness in the simple vectorization, GACM also regularizes the contour corners. Additionally, given that the frame field inherently encompasses information on the greenhouse edges, GACM can compensate for the loss of edge information due to network prediction errors

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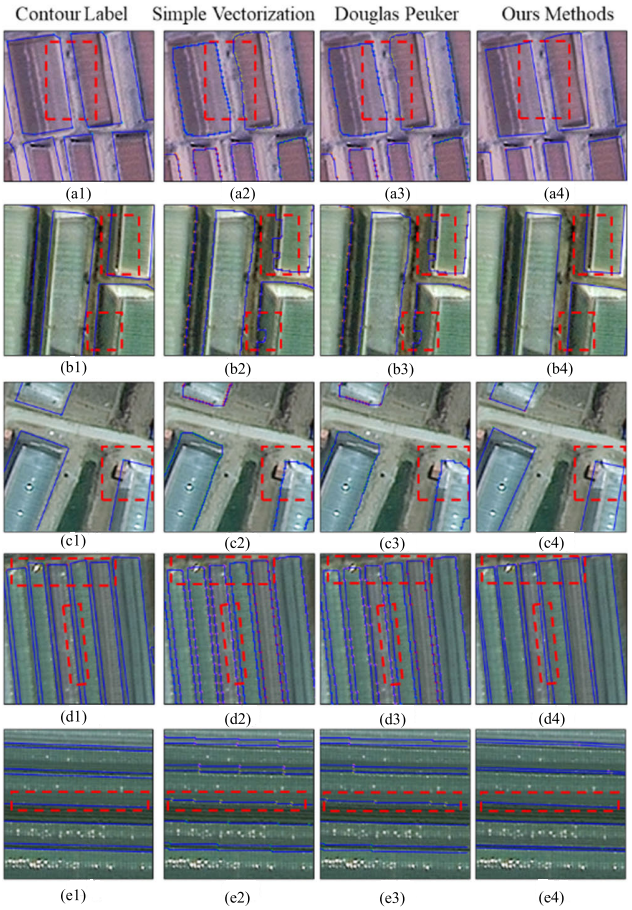


Fig. 6. Vector maps of greenhouses obtained by different vectorization methods. (a1)–(a4) Greenhouses with high orientation variability. (b1)–(b4) Greenhouses during different seasons. (c1)–(c4) Various greenhouse types. (d1)–(d4) Greenhouses with varying distribution densities. (e1)–(e4) Greenhouses with complex spatial arrangements.

[Fig. 6(b)] and rectifying incorrect greenhouse edges [Fig. 6(a) and (c)]. This results in vector greenhouse contours that are more regular and accurate. Fig. 6(d) and (e) demonstrate that GACM operates proficiently even in situations of high greenhouse density, highlighting its robust performance.

Table IV displays the time complexity and space complexity of the GACM and compares it with the DP algorithm. The time and space complexities of both algorithms are directly related to the number of contour points, nn, to be optimized. Specifically, GACM shows a lower demand in time complexity compared to the DP algorithm. However, when the number of contour points involved in the optimization is small, the execution efficiency of GACM is significantly affected by the frequency of system input/output (I/O) operations, leading to a scenario where the overall time consumption of the algorithm may be dominated by data I/O rather than computation. As a result, GACM exhibits higher time consumption than the DP algorithm when there are fewer contour points.

In terms of memory usage, GACM exhibits higher requirements compared to the DP algorithm. This phenomenon primarily stems from GACM's need to handle more intermediate variables during execution. However, despite the higher absolute memory usage of GACM, its space complexity remains linear in relation to the number of contour points nn, similar to that of the DP algorithm. This characteristic demonstrates the scalability of the GACM algorithm when dealing with large datasets.

TABLE IV
TIME AND SPACE COMPLEXITY OF GACM

	Baseline	**DP**
Time Complexity	O(n)	O(n*log(n))
Space Complexity	O(n)	O(n)
Time (per 100 nodes)	1.45 s	0.167 s
Memory (per 100 nodes)	1.83MB	1.36MB

D. Ablation Analysis

This section mainly focuses on the impact of the feature extraction backbone network and the SPP module on greenhouse segmentation performance, as well as the effect of image morphological operations on the vectorization results.

1) *Impact of Backbone Network:* Compared to using BoTNet as the backbone, this study explores a model using the UNet encoder as the feature extraction module and performs a sensitivity analysis. As shown in Table V, the lack of BoTNet backbone leads to a decrease in IoU, recall, and accuracy by 0.98%, 0.47%, and 0.47%, respectively. In addition, the lack of BoTNet leads to a 29.23% performance degradation in predicting the accuracy of the frame field.
2) *Effect of SPP Module:* SPP can standardize features of different dimensions and effectively capture the spatial hierarchy of the image. We replaced the SPP with a dual-layer convolutional network as the prediction head to evaluate the difference in accuracy. As shown in Table V, without the SPP, the model's IOU, Recall, and Accuracy decreased by 1.57%, 1.08%, and 0.75%, respectively, and the accuracy of predicting the frame field decreased by 31.34%.
3) *Impact of Morphological Opening Operation:* When the segmentation algorithm incorrectly marks a small part of the background as part of the greenhouse, directly vectorizing it can lead to the generation of inflection points, resulting in significant discrepancies [Fig. 7(a2) and (d2)]. The morphological opening can reduce noise in the greenhouse segmentation results, avoiding erroneous edges that may be introduced during the vectorization process [Fig. 7(b) and (e)]. Even in cases of minor deviations, the opening operation can smooth the extracted edges, reducing inflection points and simplifying the task of corner detection in subsequent algorithms [Fig. 7(c)].
4) *Impact of Line Simplification Operation:* Line simplification is also a key step in GACM. Redundant points in lines can lead to error accumulation in traditional ACMs during the line optimization process, possibly creating dense, jagged structures [Fig. 8(a2)] or even causing contour distortions [Fig. 8(b2) and (d2)]. GACM uses the DP algorithm for each contour inflection point. During the iteration process, this simplification may cause a loss of precision, which we rectify through a prior assumption (i.e., the contours of greenhouses

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TABLE V
IMPACT OF REMOVING VARIOUS MODULES ON NEURAL NETWORK PERFORMANCE

Metric	Baseline	Without BoTNet	Without SPP
IoU	**0.9579**	0.9485	0.9429
Accuracy	**0.9800**	0.9753	0.9726
Recall	**0.9776**	0.9730	0.9670
F_{error}	**217.39**	153.84	149.25

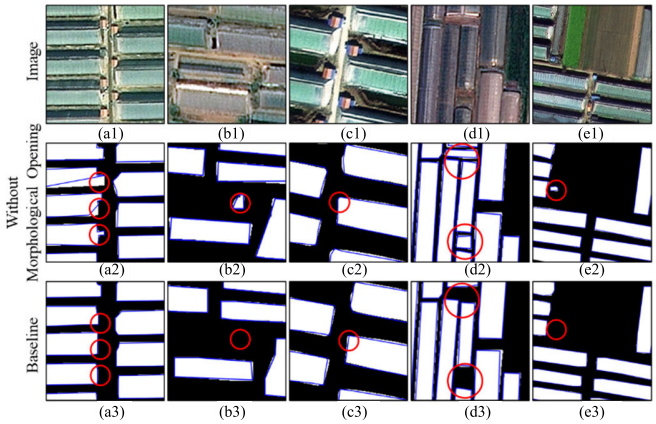


Fig. 7. Comparative results of the GACM model with and without morphological operations. (a1)–(e1) Original images of various types of greenhouses. (a2)–(e2) Segmentation results without applying morphological opening ("Without Morphological Opening"). (a3)–(e3) Segmentation results of the complete GACM model ("Baseline").

in images are rectangular) as shown in Fig. 8(b3) and (d3). However, if line simplification is not used and prior assumptions are directly applied, redundant points may lead to deformation of the contour edges [Fig. 8(c) and (e)].

V. DISCUSSION

The primary focus of this research is on the vectorization of geospatial segmentation results facilitated by the GACM, with the loss function playing a pivotal role in guiding the evolution of the GACM. Consequently, in this section, we meticulously analyze how the loss function influences the variations in greenhouse contours, as well as the impact of both the loss function and the precision of the frame field on the vectorization results.

A. How the Loss Function Influences the Variations in Greenhouse Contours?

The internal loss function plays a crucial role in driving contour changes, while the external loss function imposes constraints to ensure that the contour deviations remain minimal. In traditional ACMs, the internal loss function acts to perpetually expand the contours outward until they encounter the restrictions imposed by the external loss function. Given that the initial state in this study is already relatively accurate, the role of the internal loss function is to drive the turning of edges. As illustrated in Fig. 9(a), we take two adjacent points on the greenhouse contour as an operational object, using the midpoint of this line segment as the center of rotation to adjust the direction in situ to align with the frame field. The advantage of this approach is that when there are many redundant points on the greenhouse contour, it prevents significant deviations of the contour from its original position due to rotation. However, a drawback becomes apparent in the later stages of GACM execution when a substantial number of redundant points have been deleted. The distance between each pair of remaining adjacent points becomes relatively large, and even slight turns can lead to significant boundary changes. Fig. 10 demonstrates the scenarios in which individual loss functions are utilized separately within GACM. Under the guidance of the frame field alone [Fig. 10(a1)–(d1)], since the frame field direction away from the greenhouse edge does not point toward the greenhouse edge, it results in unpredictable final contour changes, leading to substantial contour deformations.

To address this issue, we introduce the angle internal loss function. This approach, inspired by Zhu et al., not only utilizes data to drive the model but also employs expert prior knowledge to optimize the model [37]. As depicted in Fig. 9, the angle internal loss function considers three adjacent points on the greenhouse contour as an operational object, with the middle point as the vertex. Initially, the model needs to determine whether the angle formed with the vertex at its center is closer to 90° or 180°. If the point is identified as a right angle, it might be crucial in defining the position of the greenhouse's contour. Hence, this point should be fixed while the other two points are moved to make the angle approaches 90°. If the angle is closer to 180°, all three points are moved simultaneously to align them along a straight line, preventing substantial changes in direction. It is evident that in the early stages of vector boundary optimization, the presence of excessive jagged edges makes this function less applicable. As illustrated in Fig. 10, many redundant points are identified as corner points of the greenhouse, resulting in a significant amplification of the original jagged edges on the greenhouse edge and, consequently a contour deformation.

In light of the advantages and disadvantages of the aforementioned internal loss functions, we initially employ the frame field's internal loss function. During the early iterations, the frame field internal loss function plays a pivotal role in eliminating a considerable amount of redundancy from the greenhouse contours. Once the execution of the frame field internal loss function ceases, the angle internal loss function is subsequently utilized to refine further and regularize the contours of the greenhouse. By strategically combining these two internal loss functions, this study leverages their respective strengths while mitigating their weaknesses. The frame field internal loss function swiftly addresses significant irregularities and redundancy in the initial greenhouse contours, setting the stage for the angle internal loss function to fine-tune and regularize the contours, ensuring a smoother and more accurate result.

The external loss function in GACM plays a continuous role throughout the entire iteration process, restricting the alterations that internal loss functions can impose on the contour. A binary boundary map processed through dilation is chosen as the external constraint in this research, ensuring

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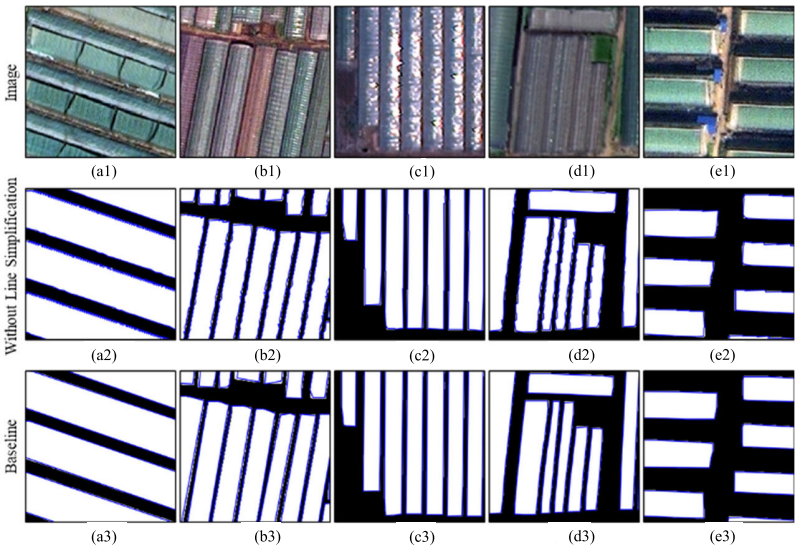


Fig. 8. Comparative results of the GACM model after the removal of line simplification operations. (a1)–(e1) Original images of various types of greenhouses. (a2)–(e2) Segmentation results without applying line simplification operations ("Without Line Simplification"). (a3)–(e3) Segmentation results of the complete GACM model ("Baseline").

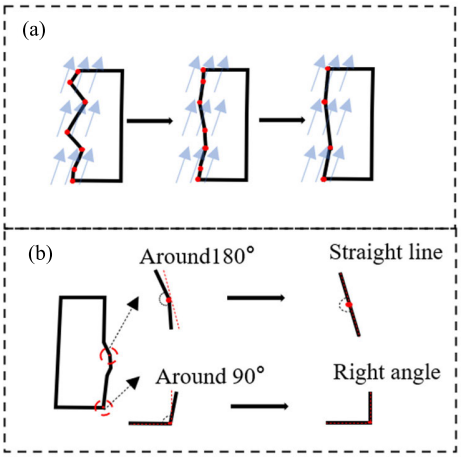


Fig. 9. Internal loss function of the GACM can drive the greenhouse contour to change. (a) Shows the effect of converging the greenhouse boundary to a smooth one guided by the frame field and (b) shows the effect of driving all the corners on the edges of the greenhouse to converge to either 90° or 180° based on a priori knowledge.

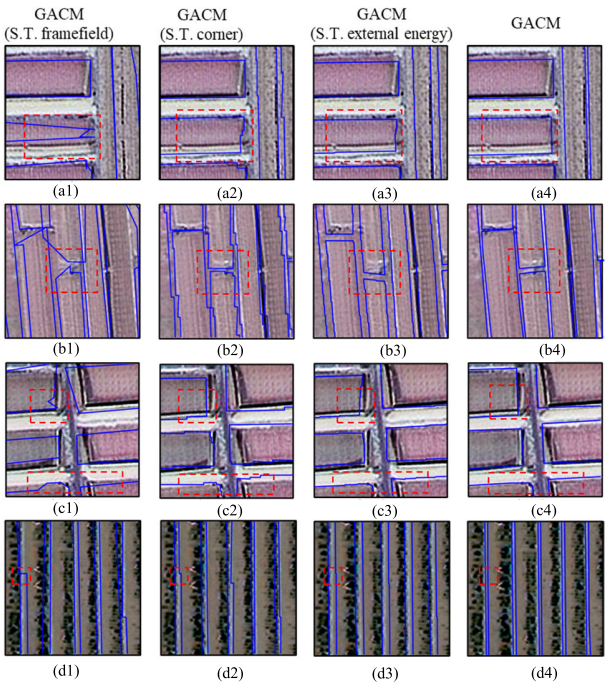


Fig. 10. Comparison of vectorization performance of GACM with different constraints. (a1)–(d1) GACM (S.T. framefield), indicating that only the framefield is used as the internal energy to drive contour modifications during optimization. (a2)–(d2) GACM (S.T. corner), which refers to adjusting the contour angles to approximate right angles or straight lines. (a3)–(d3) GACM (S.T. external energy), where only the binary segmentation results outputted by the deep learning network are used as constraints, ensuring that contour changes do not exceed the scope of the segmentation results. (a4)–(d4) GACM with all aforementioned constraints, serving as the baseline for the vectorization process in this study.

that all points within the GACM optimization process only vary within a specified range. Specifically, when a point on the greenhouse contour corresponds to a value of 1 in the predicted boundary map of the greenhouse, no action is taken; if it is 0, then the point is moved toward the nearest region with a value of 1. The rationale behind employing dilation is to allow the greenhouse to contour some space to vary under the influence of internal loss functions. The external loss function ensures that the greenhouse contour does not deform excessively after multiple iterations under the influence of the frame field internal loss function. It also limits the extent to which the angle internal loss function can deviate from the initial position when optimizing straight lines. However, the external loss function does not actively drive contour changes [Fig. 10(a3)–(d3)]. With internal loss driving forces only, GACM exhibits almost no effect, with negligible changes in IoU, recall, and MAE.

In summary, the sole utilization of any single loss function proves insufficient for achieving the smoothing and regularization effects on the greenhouse contour as facilitated by GACM.

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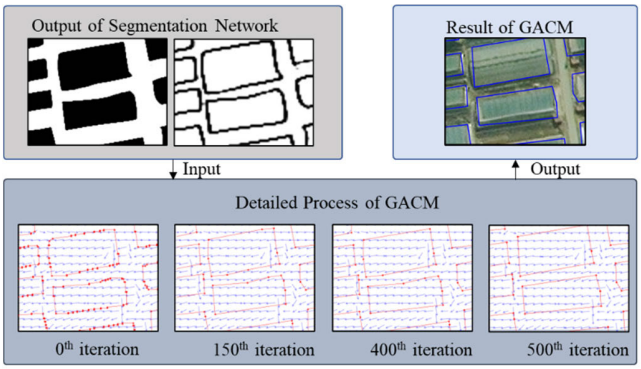


Fig. 11. Whole process of GACM processing raster data to get vector data. The inputs are the greenhouse segmentation map, edge map, and frame field map obtained from the neural network, and the regular greenhouse edges are obtained after the optimization of the boundaries of the shed by GACM.

As depicted in Fig. 11, the comprehensive process of GACM handling the greenhouse contour encompasses the following stages: First, the greenhouse binary image undergoes vectorization processing. At this juncture, the resultant vector data comprises many redundant points, with the contour presenting numerous serrations and certain edges appearing curved. Subsequently, GACM employs the frame field to rectify the greenhouse vector data, eliminating the overwhelming majority of redundant points and adjusting the contour's direction (from the zeroth iteration to the 150th iteration). Second, with the greenhouse vector contour's serrations effectively removed, GACM utilizes the angle internal loss function to drive contour deformation. Angles approximating 90° transform into right angles, while angles nearing 180° evolve into straight lines (from the 150th to the 400th iteration). Ultimately, GACM augments the weight of the angle internal loss function. Under the constraints imposed by the external loss function, it further regularizes the greenhouse contour, steering it closer to an ideal rectangle (from the 400th to the 500th iteration).

B. Impact of Frame Field Accuracy

The guidance from the frame field is a critical step for GACM to effectively eliminate the redundant points of the greenhouse (Fig. 11). Thus, the performance of the frame field maps significantly influences the performance of GACM. As depicted in Fig. 12, an error occurred in the frame field predicted by the neural network; the frame field, which should be horizontal, was predicted in a tilted direction. GACM, following this direction, guided the movement of the greenhouse contour. Since this change occurred near the boundary map, the constraint from the external energy function could not take effect, leading to an improper deformation of the greenhouse contour, which noticeably deviated from its original spatial position. However, when the frame field predicted by the network aligns well with the actual frame field, the frame field can guide GACM to correct the direction effectively, even if the binary map of the greenhouse predicted by the neural network is irregular at the edges. Therefore, ensuring that the frame field stays consistent with the actual situation is crucial in future research.

C. Impact of Imaging Conditions on Modeling Performance

The transient imaging conditions of remote sensing images lead to diversified imaging effects, which in turn affect the

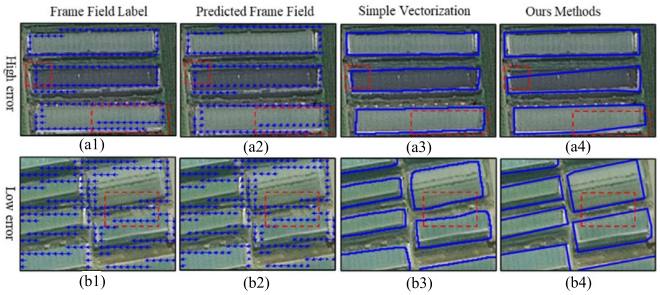


Fig. 12. Accuracy of the frame field has a significant impact on the vectorization results. (a1)–(a3) Large greenhouse vector boundary errors obtained using low accuracy frame field processing. (b1)–(b3) Small greenhouse vector boundary errors obtained using high accuracy frame field processing.

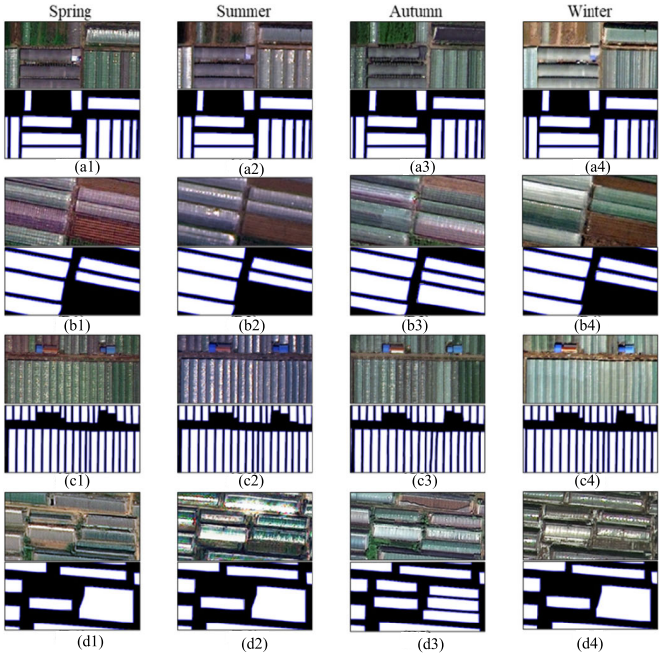


Fig. 13. Remote sensing images of greenhouses across different scenes and seasons along with the output of this study's method. (a1)–(d1) Greenhouses in spring. (a2)–(d2) Greenhouses in summer. (a3)–(d3) Greenhouses in autumn. (a4)–(d4) Greenhouses in winter. Each column represents a different scene. In each subfigure, the upper half shows the remote-sensing image of the greenhouses, while the lower half displays the output of this study's method. In the output, the blue lines are the vector boundaries produced by the GACM algorithm.

performance of deep learning models [38]. The complexity of the imaging process for greenhouses is determined by their diverse angles and the highly reflective or translucent materials of their surfaces. Particularly, the seasonal changes in the sun's elevation angle result in drastically different features of greenhouses in remote sensing images and may even cause *brightness saturation* problems in the images [Fig. 13(a2), (b2), (c2), and (d2)]. This increases the difficulty of greenhouse identification and presents challenges to the current extraction tasks.

Our proposed methods can accurately identify and extract greenhouses under different seasonal conditions (Table VI). Across the four seasons, the relative standard deviations of IoU, Recall, and Accuracy are all below 1%. Although there is a higher occurrence of brightness saturation during the summer (Fig. 13), the accuracy of the model is not significantly

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TABLE VI
PERFORMANCE OF OUR MODEL UNDER SEASONAL INFLUENCES

	IoU	Recall	Accuracy
Spring	0.9779	0.9905	0.9872
Summer	0.9925	0.9972	0.9956
Autumn	0.9766	0.9889	0.9871
Winter	0.9885	0.9960	0.9960

affected. Considering the diversity in crop types, sun elevation angles, and surrounding environments of greenhouses across different seasons, this experiment demonstrates that the method proposed in this article has a certain level of stability and reliability under varied imaging conditions.

VI. CONCLUSION

Vector data are essential for accurately representing geographic features and for vector mapping. Although progress has been made in remote sensing image segmentation using deep learning, the results often lack mask reversibility, leading to information loss and increased errors when converting these raster results to vector formats. To bridge the gap between the two formats in representing raster segmentation results, we propose a geospatial feature extraction framework based on a frame field. The framework connects the terrain segmentation process with the raster vectorization process through a trainable frame field, specifically including two modules: acquiring raster segmentation results through deep learning and vectorizing geospatial features using an ACM. A frame field connects these two processes to prevent information loss in their representations. The frame field, generated concurrently with the greenhouse raster extraction results by a three-task neural network, serves as the internal energy function in the geoscientific active process model, complemented by specially designed expert experience. Meanwhile, the edge information of the greenhouse is employed as the external energy function to constrain the deformation of the greenhouse contour, ensuring the credibility of the vectorization results.

To validate the effectiveness of the proposed method, we establish the first greenhouse extraction dataset focused on individual greenhouse objects and utilize it to train and test the framework's performance. Compared to the traditional Unet network, our neural network achieves significant improvements in metrics such as IoU, recall, and OA, particularly in the extraction performance of the frame field, with an increase of 79.44%. Regarding vectorization results, the greenhouse contours obtained using GACM show a 5.05% improvement in IoU, a 6.06% improvement in recall, and a 5.54% decrease in maximum angle error compared to those acquired through simple vectorization. Regarding the effects of smoothing and regularizing the greenhouse contours, our proposed GACM demonstrates noticeable progress compared to direct vectorization and the DP algorithm.

In future work, priority should be given to improving the accuracy of the frame field acquisition and simplifying the energy function driving paradigm during the operation of the GACM, aiming to obtain vectorization extraction results that better meet practical production needs. With the development of the remote sensing pretrained models, several potential exploration directions emerge. On one hand, the frame field, as a form of feature representation or prior knowledge for feature extraction, might already be implicitly included in the current pretrained encoding features. We need to find better ways to extract it accurately. On the other hand, considering vectors are ideal for spatial analysis, embedding the optimization process of a frame field-guided ACM into a pretrained base model could reduce errors. It could also optimize hyperparameter effects during model operations and prevent issues caused by scale changes.

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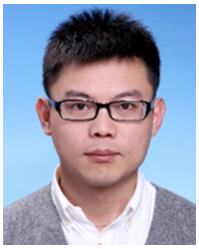
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